

Chapter 8: Working with Fractions

1. Samuel received chocolates for his birthday. He decided to share it with his family.
- He keeps $\frac{1}{3}$ rd of the chocolates for himself
 - He gives $\frac{1}{4}$ th of what is left to his sister Sarah
 - He gives the remaining 12 chocolates (except the ones which he kept for himself) to his brother

How many chocolates had Samuel received in total before he shared it?

- a) 20 b) 22 c) 24 d) None of these

Answer: c

Solution:

Samuel keeps **one-third** of the chocolates for himself.

So, the chocolates left = **two-thirds** of the total.

From this remaining amount of chocolates, he gives **one-fourth** to his sister.

The fraction of chocolates left after giving to his sister is:

Three-fourths of two-thirds

That is:

$$\left(\frac{3}{4}\right) \times \left(\frac{2}{3}\right) = \frac{1}{2}$$

So, the remaining chocolates are **half of the total chocolates**.

It is given that the remaining amount is **12 chocolates**.

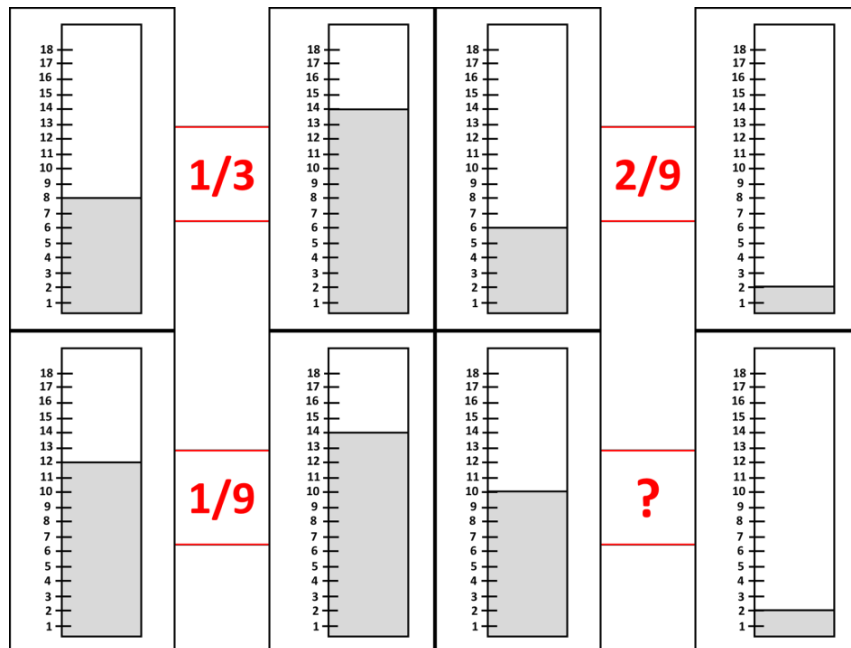
If **half** of the chocolates is 12, then the **total number of chocolates** must be:

$$12 \times 2 = 24$$

Samuel received **24 chocolates** in total.

Hence, option c is the correct answer.

2. If each of the given terms follows the same theme, what will come in place of “?”



- a) $\frac{2}{9}$ b) $\frac{2}{3}$ c) $\frac{4}{9}$ d) $\frac{1}{9}$

Answer: c

Solution:

In each group, we observe two measuring cylinders.

Each cylinder represents a fraction out of 18 (since the scale goes up to 18).

The number written between the groups shows the difference between the larger fraction and the smaller fraction, written in simplified form.

First Term:

The left cylinder shows $8/18$.

The right cylinder shows $14/18$.

Now find the difference:

$$14/18 - 8/18 = 6/18$$

$$6/18 = 1/3$$

That matches the given value in the middle term.

Question Term:

The left cylinder shows $10/18$.

The right cylinder shows $2/18$.

Now, find the difference:

$$10/18 - 2/18 = 8/18$$

$$8/18 = 4/9$$

Hence, the correct answer is option c.

3. A bus started its journey from Mumbai.

- $1/4$ th of the passengers got off the bus in Pune and 35 new passengers got on the bus

- Half of the passengers got off the bus in Goa with no new passengers getting on the bus

When the bus left from Goa, the total number of passengers was 85. How many passengers boarded the bus from Mumbai?

a) 135

b) 170

c) 180

d) Cannot be Determined

Answer: c

Solution:

Let's assume P as the number of passengers who boarded the bus from Mumbai.

$1/4$ th of the passengers got off the bus in Pune and 35 new passengers got on the bus.

Thus, the total number of passengers at Pune is $0.75P + 35$

Half of the passengers got off the bus in Goa. Thus, the total number of passengers at Goa is $(0.75P + 35) / 2 = 85$.

$$\text{Thus, } 0.75P + 35 = 85 \times 2$$

$$0.75P = 170 - 35$$

$$3/4 (P) = 135$$

$$P = 135 \times (4 / 3)$$

$$P = 180.$$

Thus, 180 passengers boarded from Mumbai.

Hence, option c is the correct answer.

4. Which of the following statements is/are sufficient to answer the given question?

Question: What fraction of the initial liquid remains in the vessel after Day 2?

Statement 1: The vessel originally contained 100 litres of liquid

Statement 2: On Day 1, the amount of liquid that evaporates is equal to half of the quantity that would remain if one-third of the initial liquid were removed

Statement 3: On Day 2, the liquid left in the vessel becomes one-fourth of the quantity that was present at the beginning of Day 2

- a) Statement 3 alone is sufficient
- b) Both Statement 1 and Statement 2 are necessarily required
- c) Both Statement 2 and Statement 3 are necessarily required
- d) Question cannot be answered even if all the Statements are used

Answer: c

Solution:

Statement 1:

It only tells us the starting quantity.

It does not tell us how much evaporates.

Statement 1 alone is insufficient.

Statement 2:

If one-third is removed, then two-thirds remains.

Half of that two-thirds is equal to one-third.

So, on Day 1, one-third of the initial liquid evaporates.

After Day 1, two-thirds remains.

But we still don't know what happens on Day 2.

Statement 2 alone is insufficient.

Statement 3:

After Day 2, only one-fourth of the Day 1 quantity remains.

But we don't know how much was left after Day 1.

Statement 3 alone is insufficient.

Now, combine Statement 2 and 3

After Day 1: two-thirds remains.

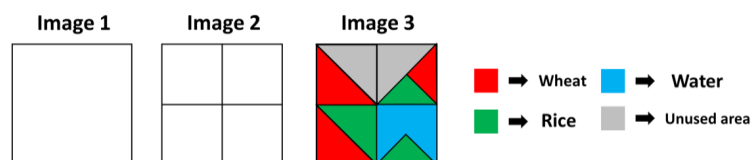
After Day 2: one-fourth of that remains.

So, final fraction = $\frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$.

Both Statement 2 and Statement 3 are necessarily required to answer the question.

Hence, the correct answer is option c.

5. A farmer has a square farm as shown in Image 1. He divides it into 4 equal parts as shown in Image 2. Now, he allotted certain portions of land for specific purposes (Rice crop, Wheat crop, Water) and some part of the farm is unused, as shown in Image 3. What fraction of the new farm is occupied by wheat?



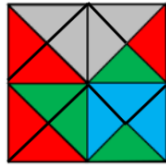
- a) $\frac{3}{16}$
- b) $\frac{1}{4}$
- c) $\frac{5}{16}$
- d) $\frac{6}{16}$

Answer: c

Solution:

The whole farm is first divided into 4 equal large squares.

Assume that each square is further divided into 4 equal triangles. So, the farm in Image 3 can be considered as:



So, total number of equal small triangles in the farm: $4 \times 4 = 16$ triangles

(This will make our calculation easier, as every portion has an equal area)

Now, count the triangles that are shaded to represent wheat.

We can see that 5 small triangles are shaded red.

Since each small triangle represents one equal part of the farm,

Fraction of the farm occupied by wheat = Number of wheat triangles / Total number of triangles = $5 / 16$

Therefore, the fraction of the farm occupied by wheat is $5/16$.

Hence, option c is the correct answer.

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6. A wire is bent to form a rectangle with length L and breadth B . After some time, the length is increased by one-fifth of its original value, and the breadth is decreased by one-fifth of its original value. After this change:

- The perimeter of the rectangle remains the same
- The area of the rectangle decreases by 100 sq. cm

What was the original length of the rectangle?

- a) 25 cm b) 40 cm c) 45 cm d) 50 cm

Answer: d

Solution:

Let the original length be L and the original breadth be B .

The new length becomes:

$$L + (1/5)L = 6L/5$$

The new breadth becomes:

$$B - (1/5)B = 4B/5$$

Since the wire is not cut or stretched, the perimeter remains the same.

$$\text{So, } 2(L + B) = 2(6L/5 + 4B/5)$$

Divide both sides by 2:

$$L + B = 6L/5 + 4B/5$$

Multiply the entire equation by 5:

$$5L + 5B = 6L + 4B$$

Rearranging:

$$5L - 6L = 4B - 5B$$

$$-L = -B$$

So, $L = B$ (multiply by ' $-$ ' on both the sides)

This means the original rectangle was actually a **square**.

$$\text{Original area} = L \times B$$

Since $L = B$,

$$\text{Original area} = L^2$$

$$\text{New area} = (6L/5) \times (4L/5) = 24L^2/25$$

Decrease in area = $L^2 - 24L^2/25 = L^2/25$

Given decrease is of 100 cm^2

So, $L^2/25 = 100$

Multiply both sides by 25:

$L^2 = 2500$

$L = 50$

The original length of the rectangle was 50 cm.

So, the correct answer is option d.

7. A number sequence is mentioned below:

19/2, 38/6, 76/18, 152/54, ...

Which will be the first term of the series that will be less than 1?

a) 7th term of the series

b) 8th term of the series

c) 10th term of the series

d) None of these

Answer: a

Solution:

The next term in the given series is obtained by multiplying the numerator of the current term by 2 and the denominator of the current term by 3.

In this way, the subsequent terms are as follows:

304/162, 608/486, 1216/1458.

Since 1216/1458 will be less than 1, the 7th term is the first term in the given series to be less than 1.

Hence, option a is the correct answer.

8. A box has 98 chalks. Each day, a teacher picks up a chalk to write.

When a chalk becomes 1/7 of its original size, she sets it aside.

Whenever she collects enough such small pieces to make one full chalk, she joins them and uses it. For how many days will the 98 chalks last?

a) 84

b) 98

c) 112

d) 114

Answer: d

Solution:

It is clear that 98 chalks would last 98 days. Each chalk leaves a fraction of 1/7 its size.

So, 98 such fractions remain, which would give 14 more chalks. So, $98 + 14 = 112$ days.

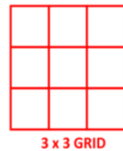
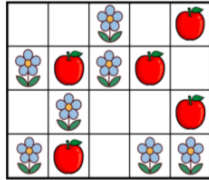
Now, the 14 chalks also leave a fraction of 1/7 its size.

14 fractions can give two more chalks.

Hence, these chalks can last 2 more days, i.e. $112 + 2 = 114$ days.

Hence, the correct answer is option d.

9. A grid contains apples and flowers in some of its cells. A 3×3 square frame can be placed on the grid without rotating and without extending outside the grid, covering exactly 9 cells. In how many different positions can the frame be placed so that at least one-third of the covered cells contain flowers?



a) 3

b) 5

c) 4

d) 6

Answer: b

Solution:

The small frame is a 3×3 grid, so it always covers $3 \times 3 = 9$ squares

The condition says at least one-third of the covered squares must contain flowers.

Now, $1/3$ of $9 = 3$

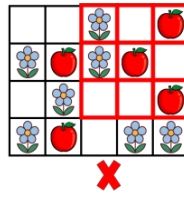
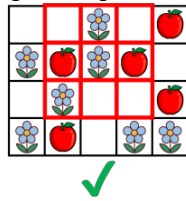
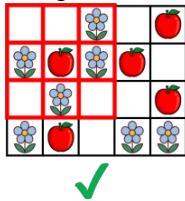
So, the frame must contain 3 or more flowers.

Any placement with fewer than 3 flowers is not allowed.

We will check the grid **column-wise**, starting from columns 1 to 3.

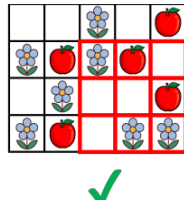
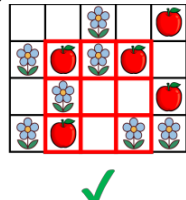
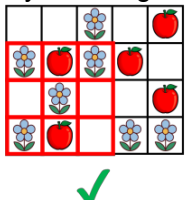
For each starting column, we place the frame over rows 1 to 3, then slide the frame to the right.

We get the following arrangements when we follow this:



Then, we move to Row 2 and repeat the same process, sliding the frame across from the first possible column to the last column of the grid.

By following this systematic row-wise and column-wise approach, we obtain these arrangements:



After checking all the possible placements, we observe that **5 arrangements** satisfy the condition of having AT LEAST one-third of the covered cells containing flowers, as indicated by the tick marks in the solution image.

Hence, the correct answer is option b.

10. Each circle is divided into equal parts, with some parts shaded. A fraction is written to the right of each circle. How many of these fractions correctly represent the shaded part of their corresponding circles?



a) 4

b) 3

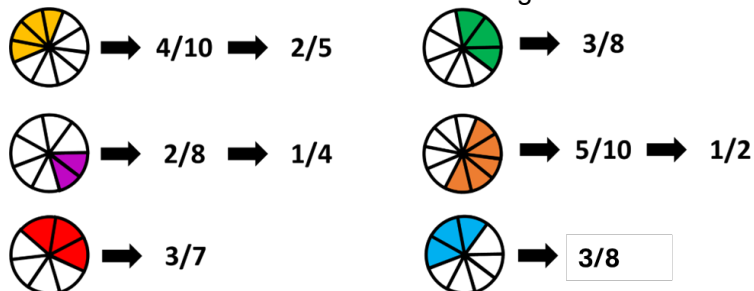
c) 2

d) 5

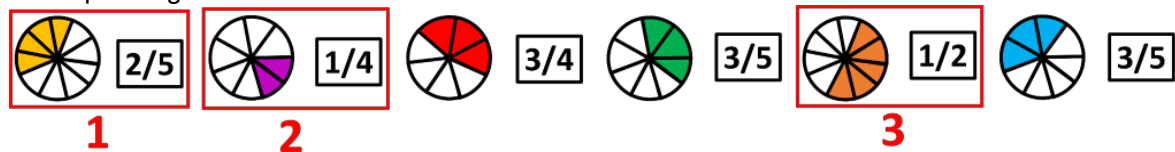
Answer: b

Solution:

The fractions of all the shaded circles are given below:



Now, we have to count the number of fractions that are correctly written to the right of their corresponding shaded circles:



As shown in the above image, a total of 3 fractions are correctly written to the right of their corresponding shaded circles.

Hence, the correct answer is option b.



The Thinking Spot

You have a collection of 10 different squares arranged as shown below. What is the minimum number of pairs of squares you need to interchange to ensure that white and black squares are arranged alternately?

Note: If you swap 1 black square with 1 white square, it is counted as 1 PAIR OF SQUARES being swapped



(a) 1

(b) 2

(c) 3

(d) 5

Answer: b

Solution:

We need to rearrange the given squares so that black and white squares are placed alternately, using the minimum number of pairwise interchanges.



The first three squares are black, followed by two white squares.

For an alternating pattern (B, W, B, W, B, W...), the 2nd position should be White, but it is Black. Similarly, position 5 should be black, but it is white.

So, we interchange 2nd Black square with the 2nd White square (position 5)



Result ■ □ ■ □ ■ ■ ■ □ □ □

Now, the first five positions follow the alternating pattern.

Now, look at the position 5, 6, and 7. All are Black, and positions 8, 9, and 10, are all White.

For alternation:

Position 6 should be White, but it is Black.

Position 9 should be Black, but it is White.

So, we interchange 6th Black square with the 9th White square.



Result ■ □ ■ □ ■ □ ■ □ ■ □

Now, all squares are arranged alternately.

So, we required a minimum of 2 pairs of squares to be interchanged to arrange the black and white squares alternately.

Hence, option b is the correct answer.

